

Revision Diff Report: v1 $\rightarrow$ v2
“Is Volatility Targeting Just Trend Following?”
Addressing HIGH-Severity Review Issues H1, H2, H4

VolPred Research System

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1 Overview

This document details all changes made from v1 (`main.tex`) to v2 (`body_v2.tex` + `main_v2.tex`) of the paper “Is Volatility Targeting Just Trend Following?” The revision addresses the top 3 HIGH-severity issues identified in the academic review (`review_v1.tex`).

Metric	v1	v2
Page count	29	~31–32
Tables	5	6 (+1 new)
Figures	2	2 (unchanged)
Bibliography entries	18	18 (unchanged count; 0 orphans remain)
Orphan references	3	0

File structure change: v2 splits the paper into `main_v2.tex` (preamble + `\input{body_v2}`) and `body_v2.tex` (all content from abstract through bibliography). The original `main.tex` is preserved unchanged.

2 Issue H1: Circular Logic in γ –TSMOM Argument

Review concern (Issues 1.1 + 7.1): Both γ and the TSMOM loading are estimated from the same return series, creating mechanical endogeneity in the $N = 22$ cross-sectional regression. The paper frames this as a “predictive relationship” without adequate acknowledgment of the identification challenge.

2.1 Changes Made

1. **[ADDED] Split-sample robustness test (Section 3.2).**

A new paragraph “Endogeneity concern and split-sample robustness” is added after the main cross-sectional results. The test estimates γ from the first half (2007–2016) and $\hat{\beta}_{\text{TSMOM}}^{\perp}$ from the second half (2017–2026) using non-overlapping data. Results:

- Split-sample Pearson $r = 0.487$ ($p = 0.021$), bootstrap 95% CI [0.114, 0.737]
- Split-sample Spearman $\rho = 0.461$ ($p = 0.031$)
- Attenuation relative to full-sample ($r = 0.564$) is acknowledged and explained as expected given reduced estimation precision

The paragraph explicitly states that this approach “reduces but does not fully eliminate the mechanical channel” and discusses why a formal IV approach is infeasible with $N = 22$ mixed-class assets.

2. **[ADDED] Panel B in Table 2 (Cross-Sectional Predictors).**

Table 2 is expanded with a new Panel B reporting split-sample results alongside the existing full-sample Panel A, making the robustness check immediately visible to reviewers.

3. **[MODIFIED] Abstract updated.**

Added one sentence to the abstract: “A split-sample robustness check—estimating γ from 2007–2016 and TSMOM loading from 2017–2026—yields $r = 0.487$ ($p = 0.021$), confirming that the relationship is not purely mechanical.”

4. **[MODIFIED] Introduction Contribution 1 updated.**

Added: “A potential endogeneity concern arises because both γ and the TSMOM loading are estimated from the same return series: a split-sample robustness check (estimating γ from 2007–2016 and TSMOM loading from 2017–2026) yields $r = 0.487$ ($p = 0.021$), confirming that the relationship survives temporal separation.”

5. **[ADDED] Limitations paragraph 5 (Section 4.4).**

New paragraph added: “Fifth, the cross-sectional regression linking γ to TSMOM loading ($N = 22$) faces an inherent endogeneity concern...Future work with a larger cross-section of futures contracts—as in Hood & Raughtigan’s 50-asset sample—could employ a more diverse set of instruments or within-asset-class variation to address this limitation.”

6. **[MODIFIED] Conclusion updated.**

Added after the full-sample correlation: “a split-sample robustness check ($r = 0.487$, $p = 0.021$) confirms this relationship survives temporal separation of the estimation windows.”

3 Issue H2: MDD Retention Lacks Statistical Test

Review concern (Issue 7.2): The central claim (90–97% MDD retention) is based on point estimates without confidence intervals. MDD is a single extreme-value statistic with high sampling variability.

3.1 Changes Made

1. **[ADDED] New Section 2.7: “MDD Retention Bootstrap Inference.”**

A complete methodological description of the block bootstrap procedure:

- 10,000 replications, block size = 252 trading days

- Formal definition of MDD retention for each bootstrap draw (new Eq. ?? in the paper)
- Reports 5th and 95th percentiles as 90% confidence interval
- Justification: block bootstrap preserves serial dependence and path-dependent nature of MDD

2. **[ADDED] New Table 6 (“Block Bootstrap Confidence Intervals for MDD Retention”).**

	SPY	50/50	DIA	QQQ	IWM
Point estimate	93%	96%	91%	90%	97%
90% CI lower	86%	90%	83%	82%	91%
90% CI upper	97%	99%	96%	95%	100%
$H_0: \leq 80\%$	Reject	Reject	Reject	Reject	Reject

All five lower bounds exceed 80%. The one-sided test rejects $H_0: \text{Retention} \leq 80\%$ at the 1% level for all five assets.

3. **[ADDED] Bootstrap interpretation paragraph (Section 3.3).**

New paragraph: “Because maximum drawdown is a single extreme-value statistic with high sampling variability, we provide formal statistical inference via block bootstrap...Even under conservative inference, at least four-fifths of VT’s drawdown protection survives TSMOM removal for every asset tested.”

4. **[MODIFIED] Abstract updated.**

Added: “Block bootstrap confidence intervals (10,000 replications, block size = 252 days) confirm that MDD retention is statistically robust, with 90% CIs excluding values below 80% for all five assets tested.”

5. **[MODIFIED] Introduction Contribution 2 updated.**

Added: “Block bootstrap inference (10,000 replications, block size = 252 days) confirms that MDD retention is statistically robust: the 90% confidence intervals are [86%, 97%] for SPY, [90%, 99%] for 50/50, [83%, 96%] for DIA, [82%, 95%] for QQQ, and [91%, 100%] for IWM, all excluding values below 80%.”

6. **[MODIFIED] Table 3 notes updated.**

Added cross-reference: “Block bootstrap 90% confidence intervals for MDD retention are reported in Table 6.”

7. **[MODIFIED] Discussion Section 4.1 updated.**

Added parenthetical: “with block bootstrap 90% CIs excluding retention below 80% for all five assets; Table 6.”

8. **[MODIFIED] Conclusion updated.**

Added: “with block bootstrap 90% confidence intervals excluding retention below 80% for all five assets.”

9. **[ADDED] Limitations paragraph 6 (Section 4.4).**

New paragraph: “Sixth, our MDD retention estimates, while supported by block bootstrap inference (Table 6), are based on five US equity assets. Extending the MDD decomposition to the full 22-asset sample and to international markets would further strengthen the generalizability of the 90–97% retention finding.”

4 Issue H4: Three Orphan Bibliography Entries

Review concern (Issues 3.1 + 10.1): Three bibliography entries—barroso2015, daniel2016, fleming2001—appear in the bibliography but are never cited in the text. These orphan references may trigger a desk reject.

4.1 Changes Made

All three orphan entries are now cited in the text at locations where they are substantively relevant. No entries were removed.

1. **[ADDED] Barroso & Santa-Clara (2015) — cited in Section 1 (Introduction).**

Location: Paragraph 2, after discussing the leverage effect connection to VT.

Citation context: “This mechanical channel is well-documented: Barroso and Santa-Clara (2015) show that volatility-scaling momentum strategies eliminates momentum crashes, precisely because the leverage effect creates a natural hedge that scales down exposure when volatility spikes after losses.”

Rationale: Barroso & Santa-Clara (2015) directly demonstrates that volatility-scaling (the core of VT) interacts with momentum through the leverage effect—precisely the paper’s central thesis. Citing them in the Introduction establishes that the VT–momentum connection has precedent.

2. **[ADDED] Daniel & Moskowitz (2016) — cited in Section 1 (Introduction, last paragraph before roadmap).**

Location: “Our paper contributes to several literatures” paragraph.

Citation context: “Our finding that VT’s embedded momentum exposure operates through the leverage effect channel complements [Daniel and Moskowitz \(2016\)](#), who show that momentum crashes are driven by precisely this leverage-effect-induced volatility clustering.”

Rationale: Daniel & Moskowitz (2016) directly links leverage-effect-induced volatility to momentum crashes, providing the theoretical bridge between VT and TSMOM that the paper exploits.

3. **[ADDED]** Fleming, Kirby & Ostdiek (2001) — cited in Section 1 (Introduction, last paragraph before roadmap).

Location: “Our paper contributes to several literatures” paragraph.

Citation context: “We also extend the ‘volatility-as-insurance’ perspective of [Lai \(2026a\)](#) to a global setting, building on the foundational work of [Fleming et al. \(2001\)](#), who established the economic value of volatility timing in a portfolio context.”

Rationale: Fleming et al. (2001) is the foundational paper on the economic value of volatility timing—the intellectual ancestor of the VT literature. Citing them grounds the paper’s contribution in the established literature.

4.2 Verification

After these additions, all 18 bibliography entries are cited at least once in the text. The orphan count drops from 3 to 0.

5 Summary of All Text Changes by Section

Section	Type	Description
Abstract	[MODIFIED]	4 sentences (split-sample result; bootstrap CIs)
1. Introduction, §2	[MODIFIED]	1 sentence citing Barroso and Santa-Clara (2015) on leverage effect and momentum crashes
1. Introduction, Contribution 1	[MODIFIED]	1 sentence on endogeneity concern and split-sample result
1. Introduction, Contribution 2	[MODIFIED]	4 sentences on bootstrap CIs with specific numbers

Section	Type	Description
1. Introduction, “contributes to”	[MODIFIED]	ED citations: Daniel and Moskowitz (2016) and Fleming et al. (2001) integrated into existing paragraph
2.7 MDD Boot-strap	[ADDED]	New subsection (methodology for block bootstrap) + new equation (Eq. ??)
3.2 Cross-section	[ADDED]	New paragraph: “Endogeneity concern and split-sample robustness” (~180 words)
Table 2	[MODIFIED]	Panel B (split-sample results); table notes expanded
3.3 Dual Mechanism	[ADDED]	New paragraph: “Bootstrap inference for MDD retention” + new Table 6
Table 3 notes	[MODIFIED]	Sentence cross-referencing Table 6
4.1 Discussion	[MODIFIED]	parenthetical referencing bootstrap CIs and Table 6
4.4 Limitations	[ADDED]	+2 new paragraphs (5th: endogeneity limitation; 6th: MDD generalizability)
5. Conclusion	[MODIFIED]	ED clauses (split-sample result; bootstrap CIs)
Bibliography	—	No entries added or removed; orphan count: 3 → 0

6 Impact Assessment

6.1 Page Count

- v1: 29 pages
- v2: ~31–32 pages (estimated +2–3 pages from new content)
- Net additions: ~600 words of text + 1 new table + 1 new equation

6.2 Claim Strength Changes

Claim	v1 Strength	v2 Strength
γ predicts TSMOM loading	“empirical regularity” (point estimate only)	“empirical regularity” + split-sample confirmation ($r = 0.487$, $p = 0.021$) + explicit endogeneity discussion
90–97% MDD retention	Point estimates, no CI	Point estimates + 90% bootstrap CIs (all lower bounds $> 80\%$) + formal H_0 test
VT–TSMOM connection via leverage effect	Supported by 3 cited papers	Supported by 3 cited papers + Barroso and Santa-Clara (2015) + Daniel and Moskowitz (2016) + Fleming et al. (2001)

6.3 Remaining HIGH Issues Not Addressed in This Revision

The review identified 8 HIGH-severity issues total. This revision addresses 3 (H1, H2, H4). The remaining 5 are:

- 5.1** BAB proxy (SPLV–SPHB) sample truncation — requires AQR BAB data download
- 8.1** Inconsistent sample periods — requires data re-analysis
- 9.1** MDD retention for all 22 assets — requires additional computation
- 3.1** Missing key literature beyond orphans (Liu et al. 2019, etc.) — partially addressed by citing the 3 orphans; additional citations deferred
- 7.1** IV approach for cross-sectional regression — acknowledged as limitation in v2; full IV deferred to future work

References

Barroso, P., & Santa-Clara, P. (2015). Momentum has its moments. *JFE*, 116(1), 111–120.

Daniel, K., & Moskowitz, T. J. (2016). Momentum crashes. *JFE*, 122(2), 221–247.

Fleming, J., Kirby, C., & Ostdiek, B. (2001). The economic value of volatility timing. *JF*, 56(1), 329–352.

Lai, Y.-H. (2026a). Leverage direction matters. *Working Paper*, Da-Yeh University.